

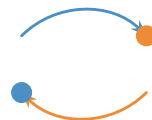
Chapter 1

Introduction

People interact constantly, and much of that interaction is strategic: one person's well-being depends on what someone else decides to do.

A firm competing for market share, two colleagues vying for the same promotion, and rival candidates in an election are all strategic settings in this sense, even though only the last one looks like a “game” in the everyday meaning of the word.

Game theory is the systematic study of this kind of interdependence.



Strategic Settings

Definition. **Interdependence** describes a situation in which one person's behavior affects another person's well-being.

Definition. A **strategic setting** is a situation of interdependence: to decide how best to behave, a person must consider how the people around him will choose to act.

Business partners deciding how to divide the work on a joint project, firms fighting for the same customers, and workers competing for one promotion are all strategic settings, whether the interaction is cooperative, competitive, or some mixture of both.

Why Study Game Theory?

A systematic theory of strategic interaction is useful for reasons that go beyond any single application.

- It gives us a common **language** for discussing and comparing strategic situations that might otherwise look unrelated.
- It gives us a **framework** for building models of strategic settings, which forces us to be precise and rigorous about what we are assuming.
- It lets us trace out the **logical implications** of an assumption about behavior, rather than relying only on intuition.

Later in the course, we will also see that game theory helps us understand how strategic forces shape economic and social outcomes, and helps individual decision makers act well within them.

A Brief History

Systematic thinking about strategic behavior is old. The Babylonian Talmud already set out rules for dividing an estate that anticipate modern theories of fair division, and mathematicians were formally analyzing parlor games centuries ago.

The modern theory, however, is a twentieth-century achievement. John von Neumann, working with Oskar Morgenstern in the 1920s through 1940s, gave the first rigorous, general mathematical treatment of strategic situations and called it the theory of games.

Their approach could handle only a narrow class of settings. It was John Nash, working at mid-century, who supplied the concepts of rational behavior, most importantly the equilibrium concept that bears his name, that let the theory address a much wider range of situations.

Noncooperative Game Theory

This course focuses mainly on **noncooperative game theory**, a framework that treats every player's action as an individual choice, made independently of the other players present.

Definition. An **individual action** is something a person decides on her own, independently of the other people in the strategic setting.

Like a sport or a board game, a noncooperative game is described by a set of rules: how many players there are, what each can do, and how the game ends.

The CPBL's rules (Taiwan's professional baseball league) specify who may be on the field and how outs are recorded; Monopoly's rules specify the order of play and how a winner is determined. Careful specification of the rules is just as important in the formal theory.

Cooperative Game Theory

Sometimes it is more useful to abstract away from the details of who proposed what to whom, and to think of the outcome of a negotiation as a single **joint action** chosen by the parties together. This alternative approach is called **cooperative game theory**.

Cooperative game theory is especially useful for studying contractual relationships, since a great deal of strategic interaction, between a worker and a manager, between a homeowner and a contractor, between two governments negotiating a tariff, is ultimately about the terms of some agreement.

This course is built mainly around the noncooperative framework, but keep in mind that it is a choice of modeling strategy, not a claim that cooperation is rare.

What Is a Game?

Everyday language treats a game as a contest with a winner and a loser. Most strategic settings are not so clean.

Example. Two managers developing a new product together face genuine interdependence: each one's effort affects the other's payoff from the project. Yet there need not be a winner and a loser.

If both managers cooperate, both are likely to benefit from the product's success; if either shirks, both may suffer.

A worker bargaining over her wage is in conflict with her employer on that dimension, but the two may well agree that the contract should include a bonus for outstanding performance, since a well-designed bonus can benefit them both.

Defining “Game”

Definition. A **game** is a formal, mathematically precise description of a strategic setting: a specification of the players, what each can do, what each knows, how their choices determine an outcome, and how each player ranks the possible outcomes.

Definition. **Game theory** is the methodology of formally studying situations of interdependence.

Conflict and cooperation coexist in most real strategic settings. Game theory gives us the tools to study both at once, using a single, mathematically precise language.

The next chapter introduces the first of two standard ways to write a game down formally: the **extensive form**.

Solution Concepts

Writing a game down formally is only half the task. Once a game is on the table, we still need a theory of how it will actually be played.

Definition. A **solution concept** is a consistent theory of how a game will be, or should be, played: a prediction about, or a prescription for, its outcome.

Solution concepts are what separate game theory from mere bookkeeping of a setting's rules: two people can agree on how to write a game down and still disagree about which one predicts its outcome.

This course builds up a sequence of solution concepts, each adding an assumption the last one left out: iterated dominance, Nash equilibrium, mixed-strategy Nash equilibrium, subgame-perfect equilibrium, Bayesian Nash equilibrium, and perfect Bayesian equilibrium.

Key Concepts

- Interdependence and strategic setting describe situations in which one person's well-being depends on another's choices.
- A game is a formal description of a strategic setting: players, actions, information, outcomes, and preferences.
- Noncooperative game theory treats every action as an individual choice; cooperative game theory instead models the outcome of negotiation as a joint action.
- A solution concept is a theory of how a game will, or should, be played; this course develops six of them in turn.